

SIYABONGA EMMANUEL HLENGWA

Senior Frontend Engineer | Full-Stack Developer

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[LinkedIn Profile](#) | [GitHub \(nerdychap\)](#) | [Portfolio \(shlengwaprofile.netlify.app\)](#)

PROFESSIONAL SUMMARY

Analytical, high-impact Senior Frontend Engineer with over 5 years of professional experience architecting high-performance web applications, real-time enterprise systems, and cross-platform mobile products. An AI-native developer who leverages cutting-edge coding agents (e.g., Claude Code, GitHub Copilot) to maximize development efficiency, accelerate technical research, and rapidly ship features. Proven track record at premium organizations like Mercedes-Benz, seamlessly bridging the gap between pixel-perfect UI/UX design (Figma) and modern full-stack architectures (React, Next.js, TypeScript, Node.js(Express)).

CORE TECHNICAL SKILLS

- **Frontend Ecosystem:** React, React Native, Next.js, JavaScript (ES6+), TypeScript, HTML5, CSS3
- **State Management & Data:** Redux, Redux-Saga, Context API
- **UI/UX & Styling:** Figma, Tailwind CSS, Material UI, Chakra UI
- **Backend & Databases:** Node.js, Express, PHP, MongoDB, Supabase, Drupal, RESTful APIs
- **Testing & QA:** Jest, Vitest, React Testing Library, Playwright
- **AI & Developer Tooling:** Claude Code, GitHub Copilot, LLM API Integration (OpenAI/Anthropic)
- **Methodologies:** Architecture Design, CI/CD, Agile/Scrum, Code Review, Curriculum Development

PROFESSIONAL EXPERIENCE

MERCEDES-BENZ | *Front-End / Full-Stack Engineer* | 11/2022 – Present

- **System Architecture & Scale:** Architected and scaled the frontend for an enterprise-level, real-time industrial plant monitoring system using React and TypeScript, expanding deployment from 3 pilot plants to over 10 manufacturing plants globally.
- **Full-Stack API Integration:** Assisted in engineering scalable backend APIs using Node.js and Express, managing heavy data throughput backed by a MongoDB database layer.
- **Performance Optimization:** Boosted page load speeds by 20% by implementing frontend optimization strategies, including component lazy loading, resource prioritization, and custom media rendering.

- **State Management & Efficiency:** Eliminated redundant, expensive operations and unnecessary component re-renders by restructuring global state using Redux and utilizing advanced React optimization techniques (*useMemo*, *useCallback*).
- **Design System Ownership:** Customized and extended Material UI to establish an intuitive, responsive design system tailored for fast-paced plant monitoring workflows.

NOROFF UNIVERSITY COLLEGE | *Front-End Development Teacher* | 10/2022 – 11/2023

- **Curriculum Modernization:** Modernized the core frontend engineering curriculum to keep pace with industry shifts, introducing structured modules on Modern React, TypeScript, Tailwind CSS, and Redux.
- **Technical Mentorship:** Delivered online tutorials, managed project evaluations, and provided 1-on-1 engineering mentorship to elevate code quality and architectural understanding for software students.

DOTMODUS | *Front-End Engineer* | 01/2021 – 10/2022

- **Cross-Platform Delivery:** Spearheaded the end-to-end development of robust web and mobile applications across Banking, Financial Services, Law Enforcement, and internal enterprise verticals using React and React Native.
- **UI/UX Prototyping:** Owned the initial product design phase, leveraging Figma to wireframe high-fidelity user experiences before writing clean, modular component code.
- **Advanced State Architectures:** Introduced Redux-Saga to handle complex, asynchronous side effects, API calls, and background data synchronization across high-security platforms.
- **Strategic Leadership:** Guided technical decision-making on component library selection (introducing Chakra UI), tooling, and overall frontend architectural approaches to minimize technical debt.

MIRUM AGENCY | *Front-End Engineer* | 11/2019 – 12/2020

- **Brand Experience Development:** Took full technical ownership of high-traffic website development and content platforms for major consumer alcohol brands under the Distell group portfolio (including Savanna, Amarula, and Bain's Whisky).
- **Full-Stack Optimization:** Specialized in backend development using PHP, Drupal, and modern Headless CMS systems to establish smooth, low-latency API interactions with the client-facing UI.

TECHNICAL PROJECTS & INNOVATION

Full-Stack & AI-Driven Applications | *Lead Architect & Developer*

- **Modern Stack Architecture:** Engineered a suite of production-ready side projects utilizing Next.js (App Router), TypeScript, and Supabase for secure authentication, real-time database subscriptions, and cloud storage.
- **AI Tooling & LLM Integration:** Integrated advanced OpenAI and Anthropic APIs to power dynamic content pipelines, autonomous workflows, and real-time AI image generation capabilities.

- **AI-Native Workflow Execution:** Developed a highly optimized personal workflow utilizing Claude Code and GitHub Copilot for advanced debugging, code refactoring, and technical exploration, drastically compressing feature delivery timelines.

EDUCATION

BSc (Eng) (Hons) in Mechatronics Engineering | 11/2018

University of Cape Town (UCT)

- **Focus areas:** Advanced Electronics, Mechanical Design, Robotics, and Embedded Software Development.

Matric (Grade 12) | 12/2013

Umkhumbi High School

- **Achievement:** Finished top of class in matriculation year.